

Formal Benchmark Report v1

Knoema Engine Phase 20

24 deterministic runs

4 scenarios x 2 approaches x 3 local model profiles

Methodology

Runs use deterministic local scoring formulas.

No external LLM API or external framework is invoked.

Artifacts can be regenerated with runner.py.

Run Matrix

- Scenario A: memory recall
- Scenario B: relationship dynamics
- Scenario C: narrative branching
- Scenario D: scalability

Metric Definitions

Recall@k, token efficiency, scalability actions/sec, branches per 100 turns.
Composite score is used only for paired sign testing.

Scenario A

A memory_recall knoema: recall 0.789, efficiency 4.662, throughput 1141.6, branches 6.65

A memory_recall naive_llm: recall 0.500, efficiency 1.000, throughput 707.6, branches 4.95

Scenario B

B relationship_dynamics knoema: recall 0.786, efficiency 4.482, throughput 1053.8, branches 7.84

B relationship_dynamics naive_llm: recall 0.495, efficiency 1.000, throughput 631.1, branches 5.83

Scenario C

C narrative_branching knoema: recall 0.784, efficiency 4.429, throughput 1096.0, branches 11.40

C narrative_branching naive_llm: recall 0.492, efficiency 1.000, throughput 667.2, branches 8.48

Scenario D

D scalability knoema: recall 0.780, efficiency 4.280, throughput 608.9, branches 5.97
D scalability naive_llm: recall 0.487, efficiency 1.000, throughput 311.4, branches 4.44

Memory Recall Figure

Source figure: results/figures/memory_recall.svg

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B relationship_dynamics knoema: recall 0.786, efficiency 4.482, throughput 1053.8, branches 7.84

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Token Efficiency Figure

Source figure: results/figures/token_efficiency.svg

C narrative_branching knoema: recall 0.784, efficiency 4.429, throughput 1096.0, branches 11.40

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D scalability knoema: recall 0.780, efficiency 4.280, throughput 608.9, branches 5.97

D scalability naive_llm: recall 0.487, efficiency 1.000, throughput 311.4, branches 4.44

Scalability Figure

Source figure: results/figures/scalability.svg

Agent counts: 5, 10, 25, 50.

Narrative Branching Figure

Source figure: results/figures/branching.svg

Branching is normalized per 100 turns.

Significance

Paired sign-test p-value: 0.000244

All paired scenario/model profiles favor the Knoema memory policy.

Raw Artifacts

results/raw.jsonl has 24 rows.

results/summary.md contains aggregate tables.

results/figures contains 4 SVG figures.

Naive Baseline

Naive LLM receives full-history prompt context every turn.
This is intentionally transparent and token-heavy.

Mesa Reference

Mesa is documented as a capability boundary.

No Mesa performance numbers are claimed in this report.

Concordia Reference

Concordia is external and not vendored.

Use `baselines/concordia_reference.md` for comparison discipline.

Reproducibility

The report avoids wall-clock measurements in committed raw artifacts.

The same source inputs regenerate the same JSONL and summary.

Limitations

Metrics are deterministic proxies, not field trial results.
External baselines require separately executed adapters.

Next Steps

Add external adapter runs when equivalent scenarios are available.

Fold Phase 19 50-agent log replay into the next report revision.